Reg. No. | | | | | | | | | | |

[14CE81]

EIGHTH SEMESTER B.TECH. EXAMINATION, JUNE - 2019

PAVEMENT ENGINEERING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least **TWO** from each part.
2. Missing data may be suitably assumed.
3. Use of IRC code books permitted.

PART - A

1. a. Define a pavement and explain the difference between an highway and air Field pavement **08**

   b. Explain briefly the various parameters that are to be considered in the design of pavements. **06**

2. a. What is modulus of subgrade reaction. Draw a neat and detailed sketch for experimental set up for determining it. **06**

   b. For the following data, find the CBR value if one division is equal to 2 kg for the readings on dial gauge. **08**

Plunger penetration value, mm	0	0.5	1.0	1.5	2.0	2.5	3.0	4.0	5.0	7.5	10.0
Load dial readings, divisions	9	17	24	31	36	40	45	51	56	61	67

3. a. Explain the various approaches of pavement design by Empirical , Semi-empirical and theoretical approaches. **08**

   b. What is the meaning of cumulative standard axle and how its effect is considered in the design of flexible pavements? Explain with relevant equations. **06**

4. a. Explain with sketches the use of different types of joints in C C pavements. **07**

   b. What are the advantages of concrete roads and indicate how the future of concrete roads is good in India. **07**

PART - B

5. a. Explain the basic difference between the structural and functional evaluation of pavements. Name one method to evaluate both. **07**

   b. Explain the principle of BBD method of pavement evaluation. **07**

6. a. What is an overlay and why it is provided. Explain its classification. **07**

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	b.	Discuss about any one type of compacting equipment with regards to working principle, advantages and limitations.	**07**
7.	**a.**	Explain what is PPP in highway projects and how it is useful in present day context.	**07**
	b.	What is Risk analysis in transportation projects and how it is overcome, explain briefly with an example.	**07**
8.	**a.**	Explain the various preventive measures for accidents during road construction, maintenance and repairs.	**07**
	b.	Explain the use of the following with a neat sketch	**07**
		i. Signages	
		ii. Guide posts	
		iii. Guard rails	
		iv. Barriers	